EDU

AI-Powered College Decision Support Engine.

Benjamin Cooper, Julia Milliet Sciorra, Tina Zhou, Zahra Sultani, Ryan Hu.

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Problem Overview

College is one of the largest financial investments families make - a decision often made with incomplete outcome visibility.

- Students take on $100K+ debt without clear ROI
- Data exists, but is fragmented + inaccessible
- Misaligned interests → poor major choices
- Long-term financial strain
- Weak ROI outcomes

Solution

AI-powered engine translates education + labor data into clear insights

What

- Program-level ROI comparisons
- Salary vs. debt tradeoff analysis
- School + major outcome visibility

How

- Integrates federal datasets (Scorecard + BLS + FAFSA)
- LLM translates data into personalized insights

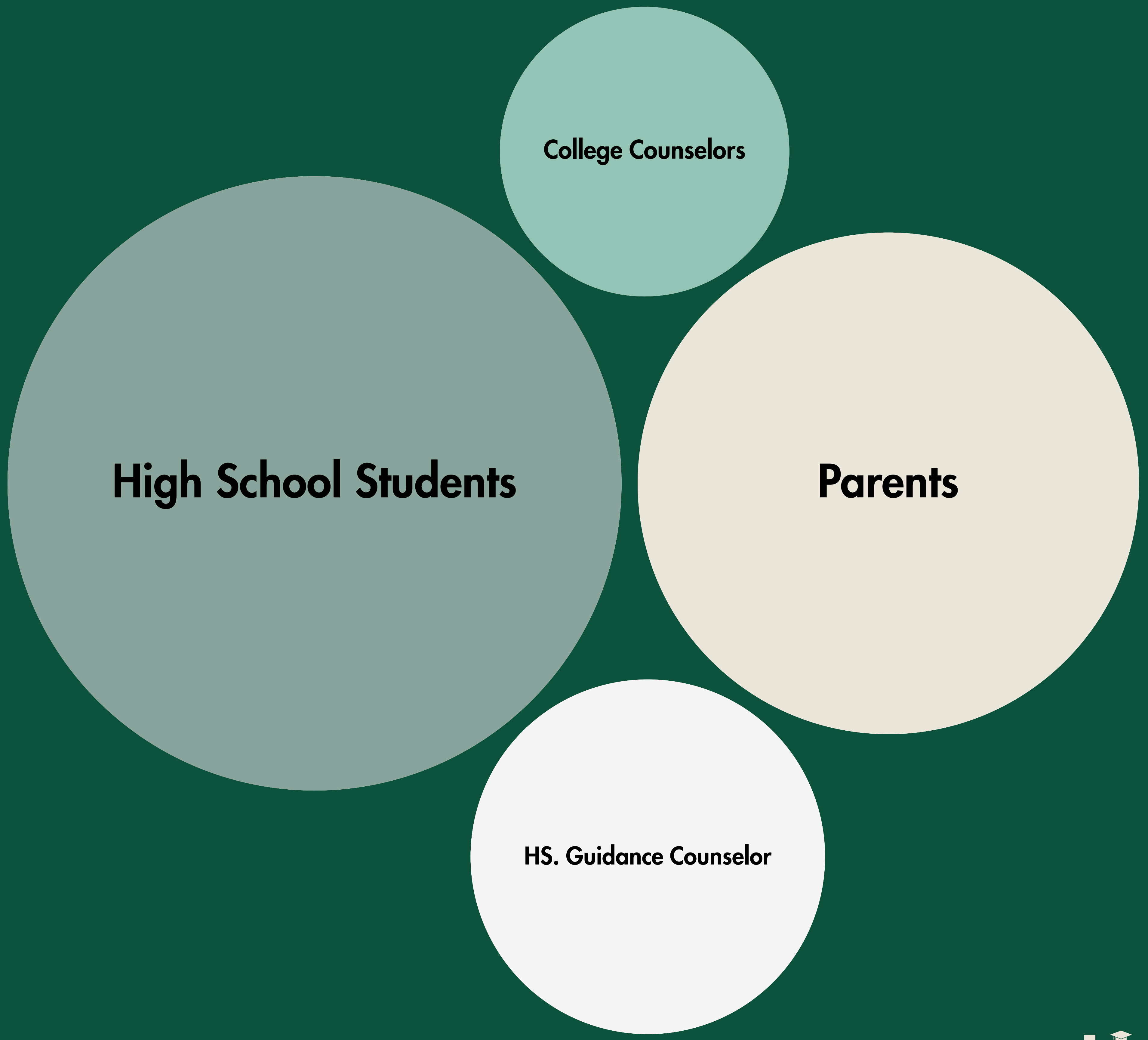
Why

- Replaces guesswork with financial clarity
- Enables confident, data-driven decisions

The Market

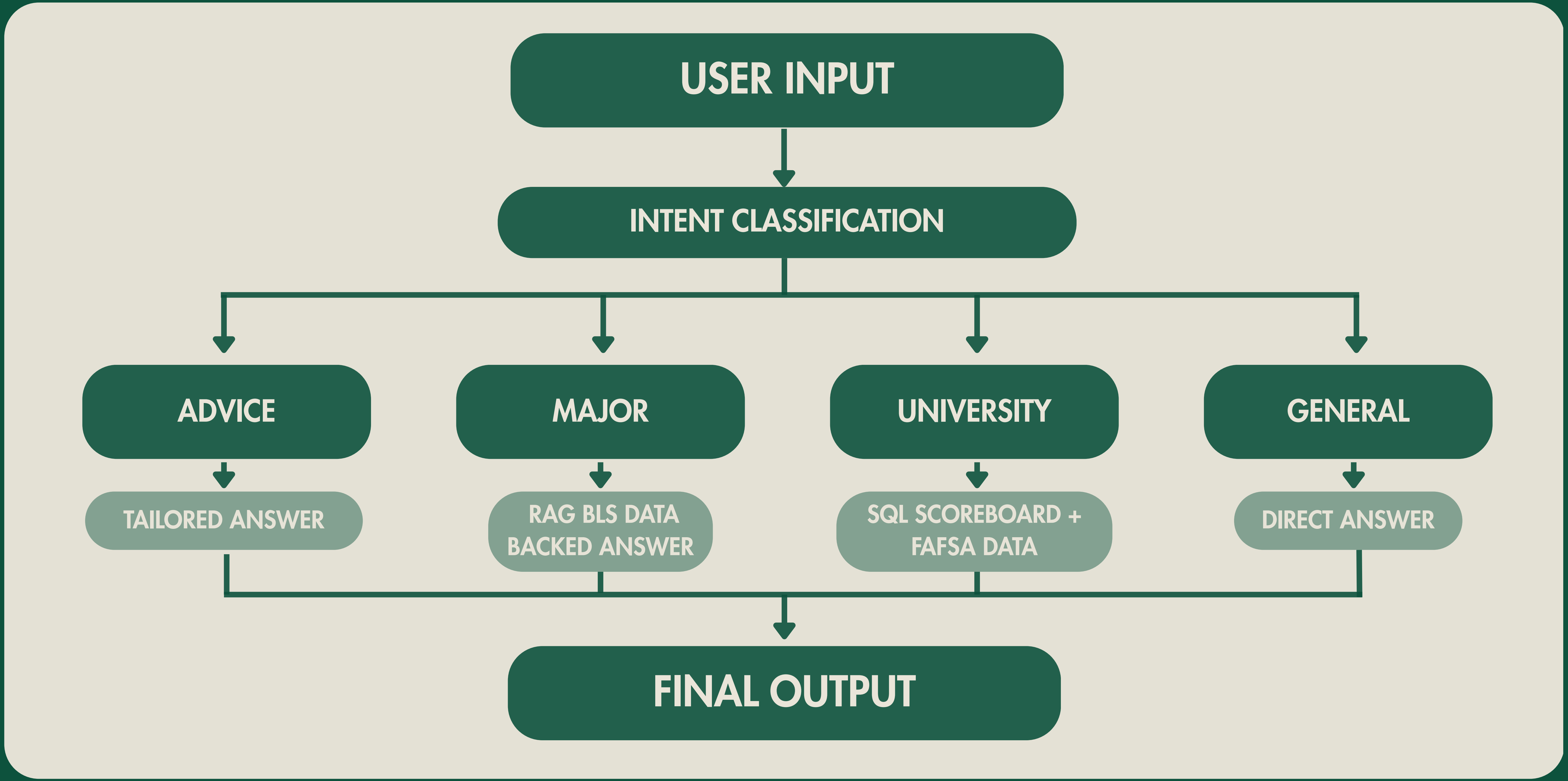
Value To Market: B2C licensing enabling scalable monetization

B2C:
- Free Version: comparisons + resume tools
- Paid Version: personalized recommendations, unlimited feature access



System Architecture

System translates user questions into data-driven answers



Data & Techniques Used

RAG + text-to-SQL enables accurate, explainable insights

- RAG → grounded answers from real BLS data
- Text-to-SQL → structured querying of College Scoreboard & FAFSA data
- Output → transparent, traceable results

Integrated federal datasets enable program-level ROI analysis

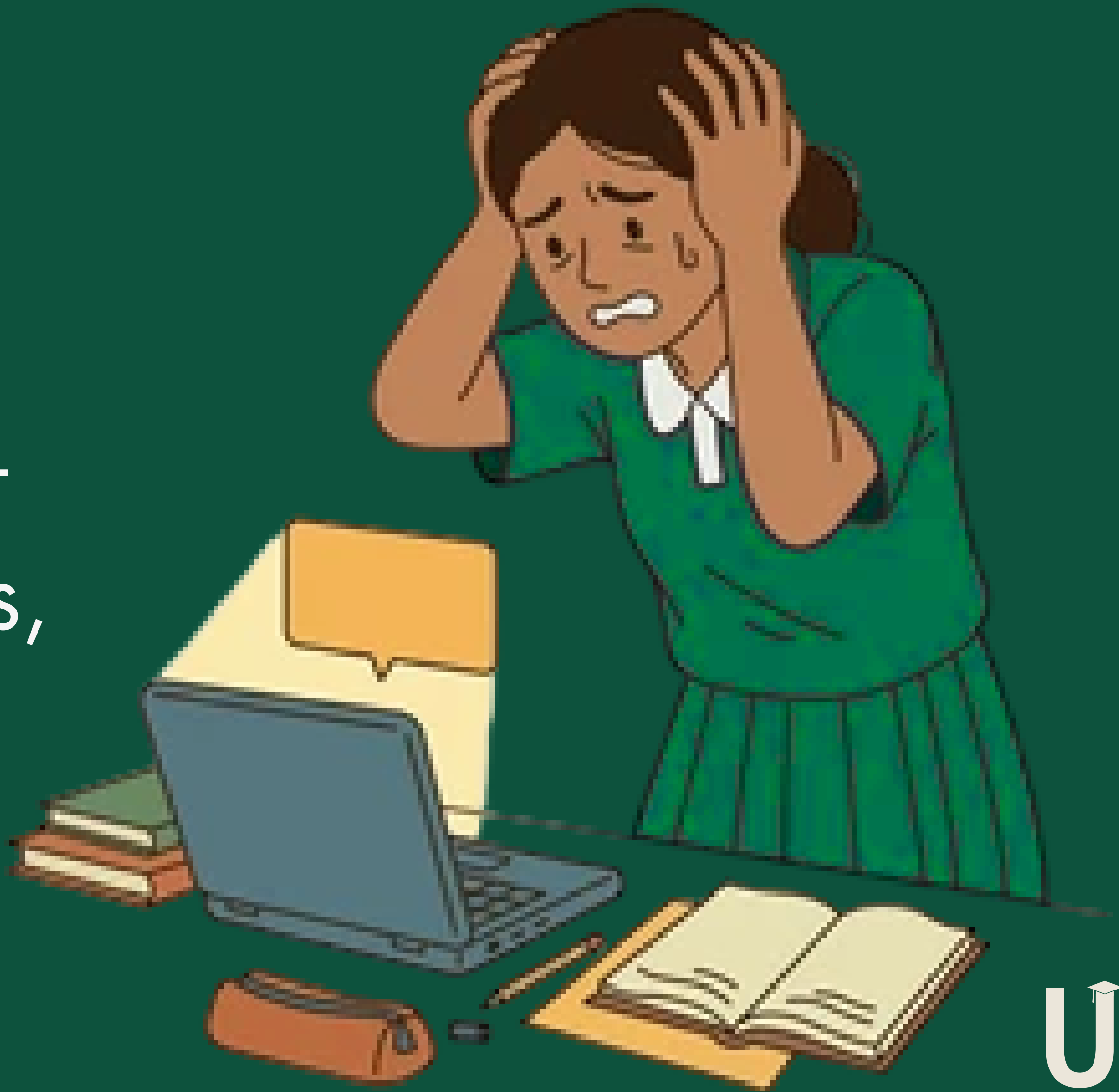
- College Scorecard → debt, graduation rates on a university level
- FAFSA → University Net Tuition Price by Family Income
- BLS → wages, employment outcomes on a major level
- Output → Layered data reveals full impact of major + university choices

Student Profile & Demo

Meet a student making a $100K+ college decision:

- Meet Jane:
    - High school senior
    - Home State: Virginia
    - SAT: 1350, GPA: 3.8/4.0

- Jane's Predicament:
    - She's always liked business and math, has thought about business analytics and supply chain degrees, but she wants to explore other options.



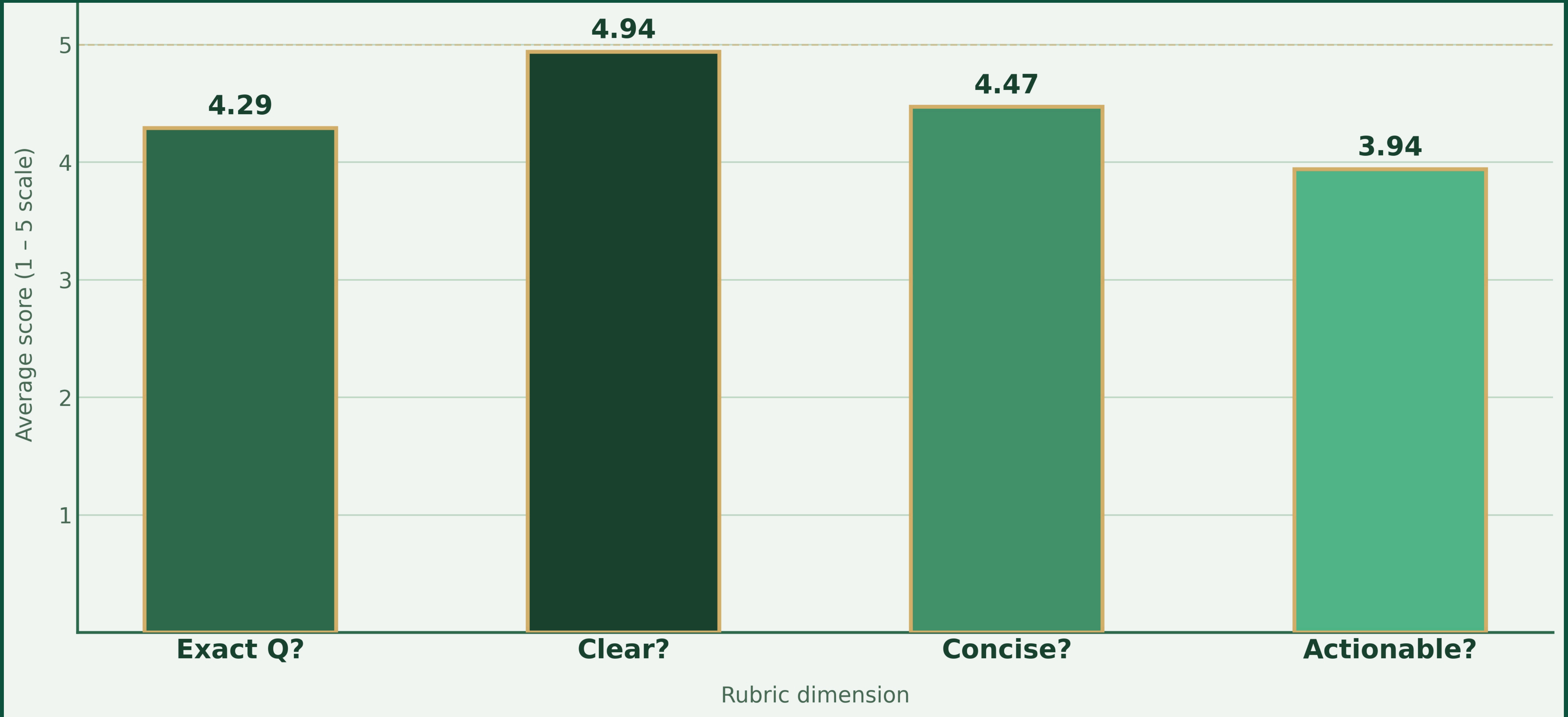
Validation Criteria

Two-Part Evaluation:

First, assess whether response includes correct core statistic or misinterprets question.
If correct, then evaluate the response using LLM-as-a-Judge rubric below:

Score	Exact Q?	Clear?	Concise?	Actionable?
5 ★★★★★	Fully answers what was asked, with little to no drift	Very easy to follow and well organized	Short, focused, and says just enough	Gives a genuinely useful takeaway or next step
4 ★★★★	Mostly answers the question, with only minor drift or extra detail	Generally easy to read and logically structured	A bit wordy or a bit brief, but still well controlled	Offers some helpful guidance, even if not especially strong
3 ★★★	Gets general idea of question, but misses part of what was asked	Understandable overall, though a little awkward or scattered	Some unnecessary detail or not quite enough detail	A small takeaway is there, but it is fairly limited
2 ★★	Partly answers question, or answers other question instead	Hard to follow, weak structure, confusing phrasing	Noticeably too long, too short, or poorly balanced	Very limited practical value; little guidance for the user
1 ★	Does not really answer the question asked	Confusing or difficult to interpret	Far from the right level of detail	No real takeaway or next step

Average Validation Scores



All dimensions score above 3.9, indicating reliable performance. Clarity leads (4.94), while actionability (3.94) is the primary area for improvement

Validation Results

Strengths

- 85% of validated responses included correct core SQL or RAG-based statistic
- Honest about missing data: model correctly flagged unavailable data
- Model accurately computed comparison statistics

Issues Improved

- Responses were not concise enough → Limited response length
- Responses scored low on actionability → Altered system prompt to push next steps

Areas for Further Improvement

Errors were driven by inconsistent school name matching & confusion between undergraduate and graduate labor outcomes → Explore technical improvements to match name and labor variations

Next Steps

Next steps focus on validation, reliability, and user adoption

- Complete data integration - College Scoreboard Data
- Conduct user testing
- Acquire privately owned data with funding
- Expand with supplement, essay, and resume editing features
- Scale to B2B model:
    - High schools, Universities, companies like Common App and Naviance

Risks and Limitations

Key risks center on data quality, model reliability, and user trust:

Data Quality	**Overreliance**
• Incomplete data • Lag in federal reporting • Inconsistent mappings between datasets	• Outputs treated as definitive • LLM misinterpretation • Reduced independent judgment
Data Privacy	**Auditability**
• Handling sensitive user inputs • Risk of storing or exposing PII	• Ensure transparent recommendations • Limited traceability to underlying data sources

Thank U

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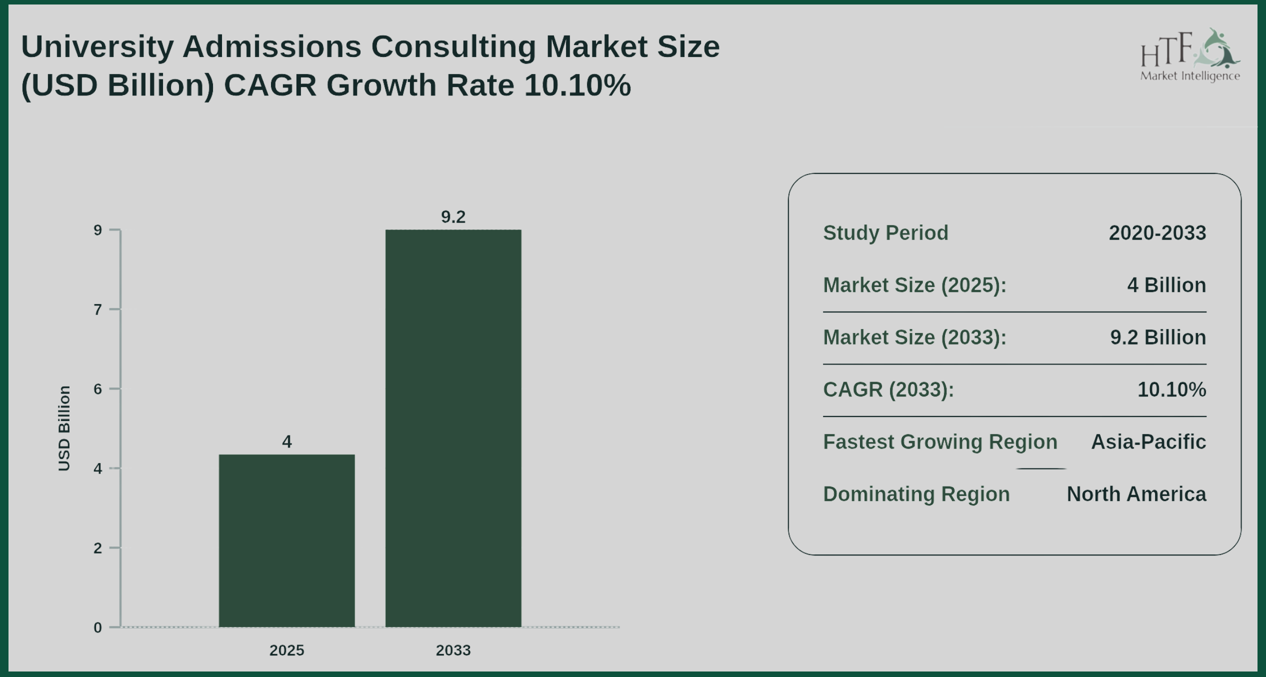
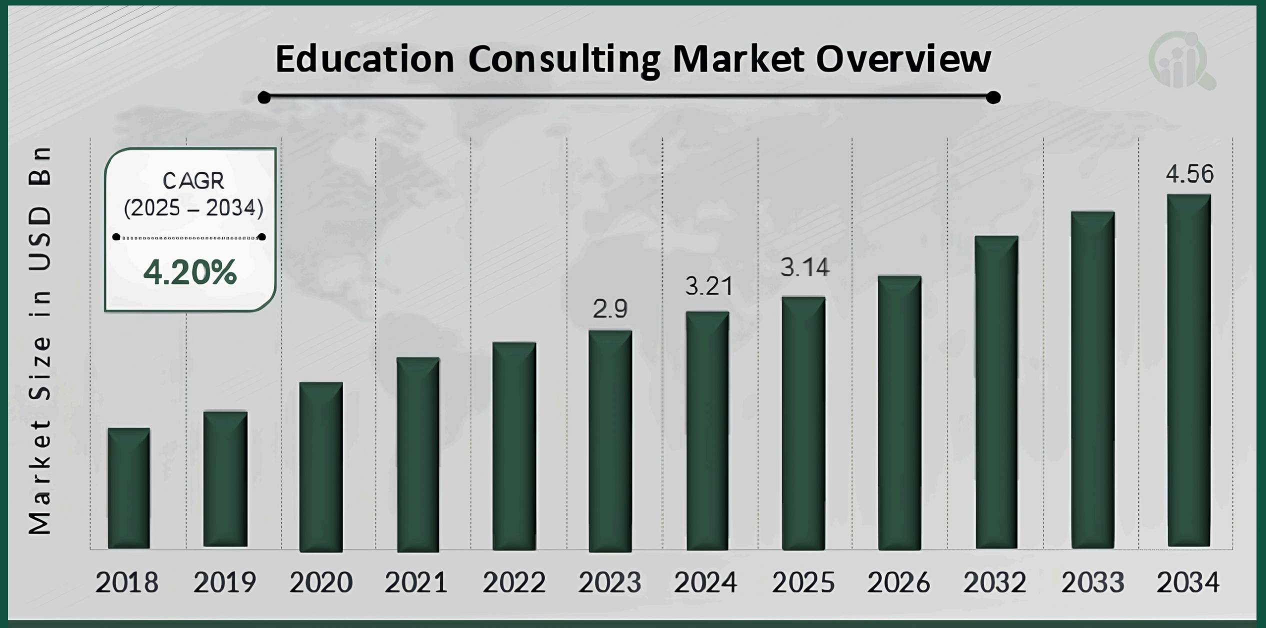
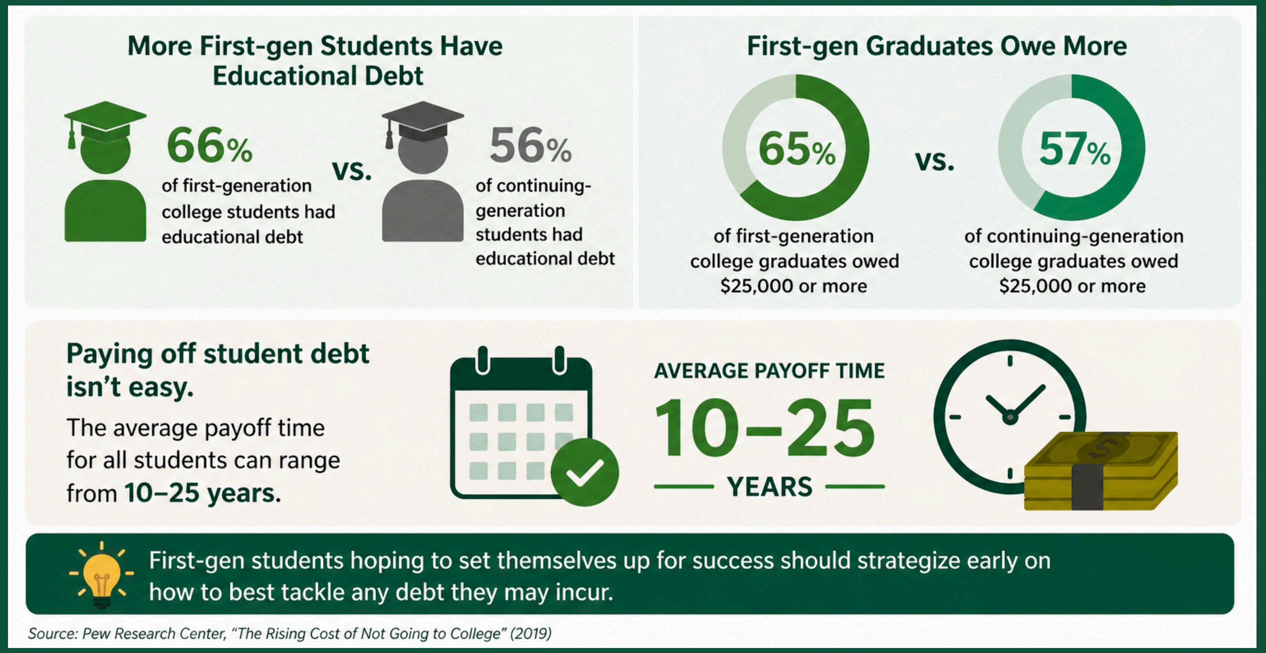
Appendices

Market Facts

- University Preparation Market
    - Global university admissions consulting market valued at ~$4.0B (2025)
    - Projected to reach ~$9.2B by 2033 (~10% CAGR)
    - Indicates sustained demand for admissions and decision-support services
- Broader Education Consulting Market
    - Global education consulting market valued at ~$3.1–$3.14B (2025)
    - Encompasses academic advising, admissions guidance, and career planning
- Market Characteristics
    - Increasing competition in university admissions
    - Growing demand for personalized guidance and decision support
    - Rising participation from international and first-generation students
    - Market is fragmented and often high-cost, limiting accessibility
- Relevance to Proposed Solution
    - Existing services focus on qualitative advising, with limited emphasis on quantitative outcomes (ROI, salary, debt)
    - Students face difficulty synthesizing:
        - Program costs, Financial aid, Post-graduation earnings
    - Our approach complements the market by:
        - Integrating public datasets (College Scorecard, BLS, FAFSA)
        - Translating data into personalized, interpretable insights
- Positions the solution within a shift toward data-driven decision-making tools in education







Validation Process

1. Codex created 20 queries
2. We manually pulled the statistics for each of the 20 queries
3. EDU was ran with those queries in separate sessions
4. If EDU's answer was correct, the response was evaluated using the LLM-as-a-Judge rubric (on slide 8)
5. Based on the results, we realized we could limit the response length to make it more concise and alter the system prompt to encourage next steps. We also found that there was a potential for error stemming from school name matching and confusion between bachelors and master's programs

Validation Sample

Question	Correct Stat	Response	Accuracy (0/1)	Exact Q? (1-5)	Clear? (1-5)	Concise? (1-5)	Actionable? (1-5)	Notes for Improvement
What is the Average Net Price Most Recent Year at George Washington University?	$51,373	George Washington University comes in at $51,373 for average net price. For the decision, weigh that against median earnings of $90,873.	1	4	5	5	4	1) The question asked only for net price — the added median earnings stat is unrequested minor drift; removing it would earn a 5 on Exact Q. 2) The 'For the decision' framing is helpful but generic; specifying what a 'good' net price benchmark looks like (e.g., vs. national average) would boost Actionable to a 5.
For Virginia Tech, what is the out-of-state tuition?	$37,764	University of the Potomac-VA Campus stands out on out-of-state tuition at $6,660, which is the lowest value in this set. If you're deciding between these options, the clearest tradeoff is University of Management and Technology for stronger earnings versus Patrick & Henry Community College for lower net price.	0	-	-	-	-	1) The query returned the lowest out-of-state tuition in a set ($6,660 for University of the Potomac-VA Campus) instead of filtering specifically to Virginia Tech ($37,764) — fix the SQL to match on 'Virginia Polytechnic Institute and State University' by name. 2) The response never mentioned Virginia Tech by name, which is a clear signal the query misfired.
Which has better bachelor's-level earnings: Economics or Business Administration?	Economics bachelor's $82,686; Business Administration, Management and Operations bachelor's $68,257; Economics is higher	Economics has higher bachelor's-level earnings. National median 4 years after completion: Economics (B.A./B.S.) $82,686 vs. Business Administration, Management and Operations (B.A./B.S.) $68,257 — about $14,400 more for Economics.	1	5	5	5	3	1) Accurate and clean. To push Actionable to a 5, add one decision-useful insight beyond the gap, which amplifies the earnings advantage over time.' 2) 'About $14,400 more' is slightly imprecise ($14,429 exact) — use the precise figure for full rigor on a data-lookup response.

#	Path	Question to Ask	Correct Stat / A	Response	Accuracy (0/1)	Exact Q? (1-5)	Clear? (1-5)	Concise? (1-5)	Actionable? (1-	Notes For improvement
1	SQL	What is the Average Net Price Most Recent Year at George Washington University?	$51,373	George Washington	1	4	5	5	4	1) The question asked only for net price — the added median ea
2	SQL	How selective is Georgetown University? What's its admission rate?	12.90%	Georgetown Univer	1	4	5	4	4	1) Response adds net price and median earnings that were not
3	SQL	If I'm looking at University of Maryland-College Park, what are median earnings 10 years after entry?	$82,860	University of Maryl	1	4	5	4	4	1) Net price and median debt are unrequested additions; answ
4	SQL	What is Howard University's median debt?	$19,200	Howard University	1	4	5	5	4	1) Median earnings ($63,066) were not asked for — while usefu
5	SQL	Compare in-state tuition at Georgia Tech and Purdue.	Georgia Tech: $12,0	East Georgia State	0	-	-	-	-	1) The SQL query returned results for the wrong institutions en
6	SQL	For Virginia Tech, what is the out-of-state tuition?	$37,764	University of the Po	0	-	-	-	-	1) The query returned the lowest out-of-state tuition in a set ($
7	SQL	What is the completion rate at American University?	78.90%	American University	1	5	5	5	3	1) The response gives the correct stat with zero drift — to impr
8	SQL	Which has the lower Average Net Price Most Recent Year, Elon University or University of Maryland-College Park?	Elon University: $4	University of Maryl	1	3	5	5	3	1) The question asked to compare both schools — the response
9	SQL	What is the Average Net Price Most Recent Year at University of Michigan-Ann Arbor?	$19,756	University of Michig	1	4	5	5	4	1) Median earnings ($83,648) was not requested — offer it as a
10	SQL	Between George Washington University and Howard University, which has higher median earnings 10 years after entry?	GW: $90,873; Howa	George Washington	1	4	5	4	4	1) The response correctly identifies GW as higher but the 'lowe
11	RAG	What are the bachelor's median earnings 4 years after completion for Industrial Engineering?	$98,442	The national media	1	5	5	5	3	1) The answer is perfectly accurate and concise but offers no t
12	RAG	How much do master's graduates in Operations Research earn 4 years after completion?	Data is not in the sc	I don't have a maste	1	5	5	4	5	1) The response correctly flags the data gap and offers bachelo
13	RAG	What is the post graduate earnings figure for Political Science and Government?	$63,293	The national media	0	-	-	-	-	1) The answer key shows $63,293 and the response states $63,
14	RAG	If I major in Education, General, what are bachelor's median earnings 4 years after completion?	$49,808	The national media	1	5	5	5	3	1) Correct and clean — Actionable score is limited because the
15	RAG	Which pays more at the bachelor's level: Economics or Business Administration, Management and Operations?	Economics: $82,68	Economics pays mo	1	5	5	5	4	1) Excellent response — all figures correct and the $14,400 gap
16	RAG	Between Industrial Engineering and Mechanical Engineering, which has higher bachelor's earnings?	Industrial Engineeri	Industrial Engineeri	1	5	5	5	4	1) Response is accurate and well-structured; to push Actionabl
17	RAG	What can I do with an Industrial Engineering degree? Compare career paths, occupations, salaries, and degree levels related t	Industrial Engineeri	Short answer: Indus Key career paths a - Operations / Man - Supply Chain & Lo - Quality & Reliabili - Analytics, Consult I only have bachelo	1	3	5	3	5	1) The answer key requires master's earnings ($119,184) — the response explicitly
18	RAG	Compare Computer Science and Industrial Engineering for earnings and common occupations.	Computer Science I	Computer Science (Common occupati If you want, I can c	1	5	5	4	5	1) All key figures are present and accurate — Concise score is a 4 because the emp
19	RAG	Compare Political Science and Government with Public Policy Analysis for career outcomes.	Political Science an	Brief comparison — - Earnings and typi - Typical career pat If you want, I can li	1	3	4	3	4	1) The answer key includes master's earnings for both majors (PolSci $78,850, Pub
20	RAG	Which has better bachelor's-level earnings: Economics or Business Administration, Management and Operations?	Economics bachelo	Economics h	1	5	5	5	4	1) Accurate and clean — nearly identical to question 15; to push Actionable to a 5,
					Avg (Accuracy=1):	**4.29411765**	**4.94117647**	**4.47058824**	**3.94117647**	

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Stress Test

Question: Based on my stats (1.1 GPA, 990 SAT), do I have a good change of getting into GWU, Howard, or UMD?

Answer: Short answer: with a 1.1 GPA and a 990 SAT your chance of admission to GWU, Howard, or UMD–College Park as a first-time freshman is very low**. Those schools are moderately to highly selective and typically admit students with much stronger academic records.**

Practical next steps: retake the SAT/ACT if you can raise scores, **but most impact will come from improving your GPA — the fastest route is starting at a community college (especially in Virginia), earn strong grades while taking math/engineering prerequisites, and then transfer after 1–2 years.** Also build relevant experience (work or internships in logistics, engineering clubs, or certifications like Lean/Six Sigma) and write a clear, honest personal statement explaining academic challenges and growth.

If you want, I **can: 1) sketch a 2-year community-college → transfer plan with courses to prepare for industrial engineering/business, or 2) list Virginia community colleges and reachable four-year schools that match your current stats. Which would you prefer?**